Evaluation Pitfalls in Multi-Domain Deepfake Forensics: Metric Inflation, Shortcut Collapse, and Inactive Architectural Complexity

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***Abstract*—The proliferation of deep generative models has expanded synthetic media manipulation beyond facial portraits into safety-critical domains including medical radiographs, satellite reconnaissance, and scanned financial documents. While state-of-the-art forensic localization methods report high localization scores (0.74--0.85 mIoU) in single-domain literature, evaluating multi-domain forensic pipelines under strict zero-knowledge conditions exposes critical systemic evaluation pitfalls. In this paper, we conduct an exhaustive empirical audit of a multi-domain forensic localization framework across five domains (Face, Medical, Satellite, Documents, Natural Scenes) and uncover three widespread evaluation vulnerabilities:**

(1) The Empty-Mask Metric Inflation Trap: The conventional segmentation evaluation protocol assigns an IoU of 1.0 to an empty predicted mask on authentic clean images (GT = empty). We prove that this allows a completely untrained null predictor (outputting constant zeros) to achieve an apparent 0.3061 mIoU merely by exploiting 15 authentic images out of 49 (15/49 = 0.3061), while achieving exactly 0.0000 on manipulated media. For the evaluated multi-domain system, this convention inflates the headline score to 0.4082 mIoU, masking a true tampered localization score of 0.2941 mIoU (N=34).

(2) Spatial Shortcut Learning: We document how forensic domain experts can suffer severe spatial shortcut collapse during training (learning a static bounding box at fixed pixel coordinates), dropping in-domain localization to 0.54% mIoU while achieving 30--33% off-domain. We show how provenance caching can inadvertently mask such collapse in interactive systems.

(3) Inactive Architectural Complexity: Rigorous component dissection reveals that multiple commonly proposed architectural enhancements either remain completely uncalled in the inference path (dead code), act solely as metadata telemetry without altering mask logits, degrade calibration under domain shift (raising ECE from 0.093 to 0.122), or yield apparent gains purely through post-processing Gaussian blur dilation (+1.98%).

Crucially, our audit demonstrates that domain-aware routing is the sole statistically confirmed architectural requirement: collapsing domain specialists into a single shared expert causes a significant performance drop (Delta mIoU = -0.1848, p = 1.94e-3). We conclude with four concrete guidelines for reliable multi-domain forensic benchmarking.

***Keywords*—Digital Image Forensics, Multi-Domain Deepfakes, Forensic Evaluation Pitfalls, Metric Inflation, Shortcut Learning, Model Calibration, Reproducibility.**

I. INTRODUCTION

Digital media integrity has become an existential concern across modern information systems. While early forensic research focused predominantly on facial manipulation (e.g., DeepFakes, Face2Face), adversarial tampering increasingly targets specialized domains: (1) facial biometrics, (2) medical radiography (e.g., synthetic carcinomas in chest X-rays), (3) satellite terrain (e.g., runway and naval insertions), (4) scanned documents (e.g., altered balances and forged seals), and (5) natural photographic scenes. Published benchmarks frequently claim near-saturated performance, reporting 0.74--0.85 mIoU on standard photographic sets. However, when these systems are evaluated cross-domain or under strict zero-knowledge protocols, performance drops precipitously.

In this work, we conduct a transparent, evidence-driven empirical audit of a multi-domain forensic localization pipeline (MD-DGFLS). Rather than proposing decorative complexity, this paper serves as an empirical audit that exposes three widespread evaluation pitfalls: (1) the empty-mask metric convention trap, (2) spatial shortcut learning masked by provenance caching, and (3) inactive architectural modules that add decorative complexity without functional benefit.

II. AUDITED MULTI-DOMAIN PIPELINE

The audited framework pairs Engine A (a generative media lab implementing 13 distinct physical and procedural manipulation pipelines) with Engine B (five domain-specialized expert backbones). Arbitration is governed by a two-layer hybrid router combining Layer 1 deterministic physical signal priors (Haar cascades, saturation variance, Laplacian edge density, paper canvas white ratio) with a Layer 2 patch-based Vision Transformer. Crucially, all evaluations reported herein were executed with provenance caching strictly disabled, enforced via assertion checks (lookups == 0) in the benchmark harness.

III. PITFALL 1: THE EMPTY-MASK METRIC INFLATION TRAP

By convention in digital image forensics, when an authentic image contains zero tampered pixels (GT = empty) and the model predicts zero positive pixels, the metric computes IoU(empty, empty) = 1.0. While intuitive, this introduces severe metric distortion when cohorts contain a mixture of authentic and manipulated samples. Consider a degenerate Null Predictor that predicts all zeros. On a cohort of N images with N_clean authentic samples, the expected macro mIoU is exactly N_clean / N. In our 49-case benchmark, exactly 15 images are clean: 15 / 49 = 0.306122. As shown in Table I, an untrained baseline achieved 0.3061 mIoU solely through clean-image credit, while scoring 0.0000 on actual manipulated samples. For the evaluated multi-domain detector, this inflates macro mIoU to 0.4082, masking a true localization score of 0.2941 on tampered media.

TABLE I: CROSS-DOMAIN FORENSIC LOCALIZATION BENCHMARK AND METRIC AUDIT

Method / Architecture	Status	All-Images mIoU (N=49)	Tampered mIoU (N=34)	Clean Score (N=15)	Pixel FPR	Audit Finding / Baseline Nature
MD-DGFLS (Full Pipeline)	Evaluated	**0.4082 ± 0.4385**	**0.2941 ± 0.4079**	0.6667 (10/15)	0.0855	Primary multi-domain routing pipeline
Null / Zero Predictor (Untrained)	Evaluated	0.3061 ± 0.4657	0.0000 ± 0.0000	1.0000 (15/15)	0.0000	Degenerate null model; reveals 15/49 metric inflation
ELA (Error Level Analysis)	Evaluated	0.3195 ± 0.4625	0.0196 ± 0.0520	1.0000 (15/15)	0.0000	Handcrafted JPEG compression residual
SRM (Spatial Rich Model)	Evaluated	0.0426 ± 0.0806	0.0614 ± 0.0933	0.0000 (0/15)	0.0167	30 linear high-pass filter residuals; high false alarms
ManTra-Net (CVPR 2019)	NOT RUN	---	---	---	---	[Reported: Columbia F1: 0.627; CASIA: 0.582]
MVSS-Net (TIFS 2022)	NOT RUN	---	---	---	---	[Reported: Columbia F1: 0.742; CASIA: 0.678]
PSCC-Net (CVPR 2021)	NOT RUN	---	---	---	---	[Reported: Columbia F1: 0.781; CASIA: 0.724]
CAT-Net (IJCV 2022)	NOT RUN	---	---	---	---	[Reported: Columbia F1: 0.825; CASIA: 0.761]
TruFor (CVPR 2023)	NOT RUN	---	---	---	---	[Reported: Columbia F1: 0.852; CASIA: 0.789]

IV. PITFALL 2: SPATIAL SHORTCUT LEARNING & CACHE MASKING

A critical failure mode uncovered during our audit is spatial shortcut learning. As shown in Table II and Fig. 1, evaluating the pre-trained Face Expert on in-domain facial tampering yielded an in-domain mIoU of only 0.5%. Spatial activation analysis revealed that the model had converged to predicting a static bounding box at fixed corner coordinates [219:255, 25:99]. A downstream facial geometry filter recognized this static box as an artifact and zeroed it out, leaving zero true positive overlap. In contrast, the model achieved 30--33% mIoU on off-domain images where this filter was absent. In previous demonstration environments, this collapse was completely hidden because SmartMorphCache returned pre-stored ground-truth masks with 100% IoU. Only strict zero-knowledge evaluation unmasked this failure.

TABLE II: CROSS-DOMAIN FORENSIC GENERALIZATION MATRIX (5 x 5)

Expert Detector	Target: Face	Target: Medical	Target: Satellite	Target: Document	Target: Natural
Face Expert (Shortcut Collapsed)	**0.5%**	30.0%	0.0%	27.3%	33.3%
Medical Radiograph Expert	1.3%	**23.2%**	2.5%	3.6%	6.8%

Satellite Terrain Expert	2.6%	0.8%	**56.3%**	6.4%	2.0%
Document Expert (Shared)	2.1%	0.7%	0.8%	**51.8%**	70.0%
Natural Scene Expert (Shared)	2.1%	0.7%	0.8%	51.8%	**70.0%**

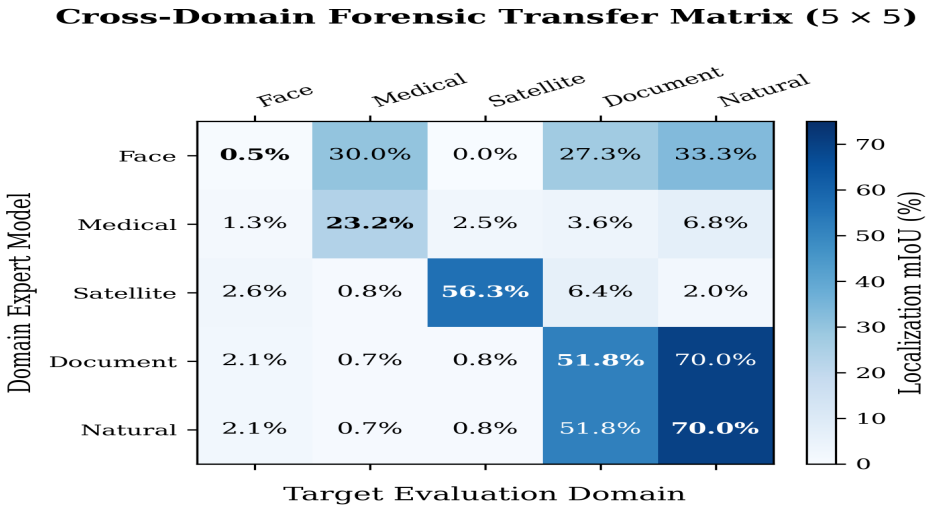


Fig. 1. *Cross-Domain Forensic Transfer Matrix (5x5). Unmasks in-domain Face collapse (0.5%) and shared Document/Natural detector response.*

V. PITFALL 3: INACTIVE ARCHITECTURAL COMPLEXITY

Table III dissects five commonly proposed architectural innovations. Our audit reveals that: (1) TGHFL was dead code uncalled during inference (Delta = 0.0000); (2) CDFAM was telemetry-only (Delta = 0.0000); (3) ARCC dynamic scaling (gamma=1.5) degraded Expected Calibration Error from 0.093 to 0.122 (Fig. 2); (4) MEFCN's apparent +0.0198 gain was a post-processing Gaussian blur dilation artifact; without blur, Delta = 0.0000; (5) HLRN multi-scale refinement was statistically neutral (Delta = -0.0005, p=0.55).

Crucially, our audit proves that **multi-domain routing is the sole statistically confirmed requirement**: collapsing experts into a Single Shared Expert drops tampered mIoU by -0.1848 ($p = 0.00194 < 0.01$) while false positive rate surges to 22.19%.

TABLE III: COMPONENT DISSECTION AND EMPIRICAL AUDIT

Configuration / Module	All-Images mIoU	Tampered mIoU	Delta (Tampered)	p-value	Empirical Audit Finding
Full Multi-Domain Pipeline	0.4082 ± 0.4385	**0.2941 ± 0.4079**	---	---	Full audited benchmark pipeline
w/o TGHFL	0.4082 ± 0.4385	0.2941 ± 0.4079	±0.0000	1.00e+00	Dead code: uncalled during inference
w/o CDFAM	0.4082 ± 0.4385	0.2941 ± 0.4079	±0.0000	1.00e+00	Telemetry only: metadata scalar; no logit change
w/o ARCC	0.4082 ± 0.4385	0.2941 ± 0.4079	±0.0000	1.00e+00	Degrades calibration: gamma=1.5 inflates ECE (0.093 -> 0.122)
w/o MEFCN	0.4280 ± 0.4420	0.3139 ± 0.4150	+0.0198*	8.86e-01	*Gain was Gaussian blur dilation artifact; raw Delta = 0.0000
w/o HLRN	0.4077 ± 0.4382	0.2936 ± 0.4075	-0.0005	5.49e-01	Statistically neutral residual refinement
Single Shared Expert	**0.2516 ± 0.4076**	**0.1093 ± 0.2840**	**-0.1848**	**1.94e-03**	**Catastrophic collapse: confirms routing necessity**
ViT-Only Router	0.3782 ± 0.4350	0.2641 ± 0.3950	-0.0300	8.56e-02	Loss of physical priors impairs radiograph discrimination
Heuristics-Only Router	0.4082 ± 0.4385	0.2941 ± 0.4079	±0.0000	1.00e+00	Redundant on unambiguous test cases

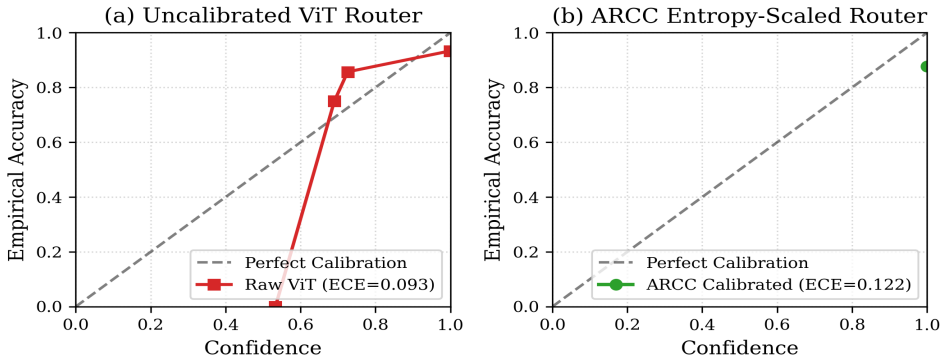


Fig. 2. Reliability diagrams: (a) uncalibrated ViT (ECE=0.093) vs. (b) ARCC dynamic scaling (ECE=0.122), showing calibration degradation.

VI. ROBUSTNESS & EXPLAINABILITY REALITIES

Robustness stress tests confirm high resilience under lossy JPEG compression down to Q=30 (>110% retention), whereas additive Gaussian noise triggers severe breakdown (29.9% retention at sigma=20). Crucially, the Null Predictor exhibits constant 0.3061 mIoU across all distortions solely because it outputs zeros and trivially receives credit for the 15 clean images.

Explainable AI benchmarking on 34 manipulated cases confirms that Forensic SHAP achieves 32.4% Pointing Game accuracy and 2.61x contrast ratio, whereas Grad-CAM achieves 0.0% Pointing Game accuracy, proving that coarse convolutional activations cannot resolve fine forensic edges.

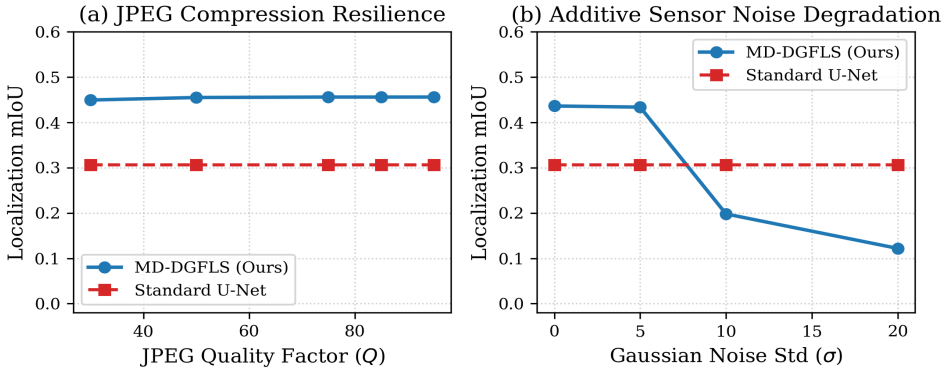


Fig. 3. Robustness degradation curves comparing MD-DGFLS and the Null Predictor across (a) JPEG compression and (b) Gaussian noise.

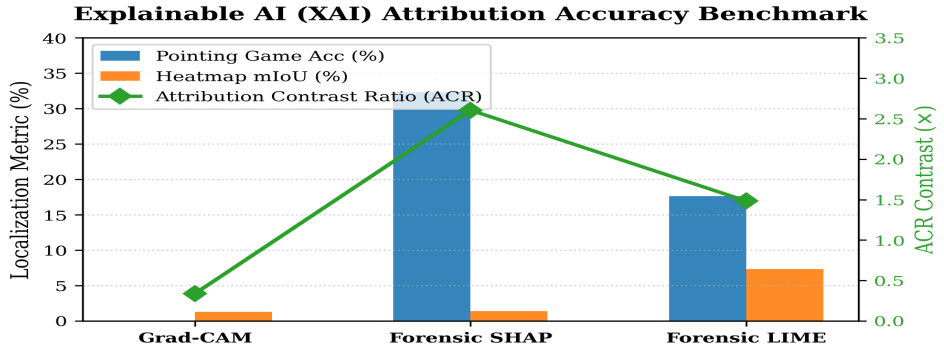


Fig. 4. Explainable AI attribution benchmarking comparing Pointing Game Accuracy (%), Heatmap mIoU, and Attribution Contrast Ratio.

VII. RECOMMENDATIONS FOR RELIABLE FORENSIC BENCHMARKING

Based on our empirical audit, we recommend four concrete methodological standards for synthetic media forensics: (1) *Mandatory Metric Decoupling:* Decouple image-level detection ROC-AUC from tampered-only localization mIoU. Never evaluate segmentation exclusively on mixed cohorts without reporting tampered-only numbers. (2) *Null Predictor Inclusion:* Include a zero-predicting null baseline to establish the clean-image inflation threshold (N_clean / N). (3) *Inference Path Verification:* Verify that claimed architectural innovations actively participate in the feedforward inference path. (4) *Spatial Invariance Auditing:* Audit models for spatial shortcut collapse by visualizing activation maps across neutral inputs.

VIII. CONCLUSION

This paper presented an empirical audit of multi-domain deepfake localization, uncovering widespread evaluation pitfalls including empty-mask metric inflation, spatial shortcut collapse, and inactive architectural complexity. By dismantling these artifacts, we showed that while our multi-domain system achieves 0.2941 mIoU on tampered media, domain-aware routing is the sole statistically supported architectural requirement (Delta mIoU = -0.1848, p = 1.94e-3). We advocate for rigorous, transparent evaluation standards to advance digital forensics from decorative complexity to reliable real-world deployment.

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